



SPARTIFICIAL

Spartificial Innovations Private Limited

Classification of Variable Stars Using Classical Machine Learning methods

Anvit Navin Khade
Girnare Vedant Santosh
Rucha Gavade

Mr. Rohan Shah

Abstract

We present an evaluation of the performance of an automated classification of the *Hipparcos* periodic variable stars into 22 types. The sub-sample with the most reliable variability types available in the literature is used to train supervised algorithms, including Random Forest (RF), Decision Trees (DT), XGBoost (XGB), Support Vector Machines (SVMs), and Multinomial Regression, to characterize the type dependencies on a number of attributes. The most useful attributes evaluated with the random forest methodology include, in decreasing order of importance, the period, the amplitude, the $V - I$ colour index, the absolute magnitude, the residual around the folded light-curve model, the magnitude distribution skewness, and the amplitude of the second harmonic of the Fourier series model relative to that of the fundamental frequency. In standard cross-validation (CV) experiments, the rate of correct classification is between 90 and 100 per cent for most of the models, depending on the variability type. The main mis-classification cases, up to a rate of about 10 per cent, arise due to confusion between SPB and ACV blue variables and between eclipsing binaries, ellipsoidal variables, and other variability types. The data used for classification was heavily imbalanced, which posed additional challenges for the supervised classification algorithms and we had to find a few ways to navigate around this issue. Our training set and the predicted types for the other *Hipparcos* periodic stars are available online.

Key words: methods: data analysis – methods: statistical – techniques: photometric – catalogues – stars: variables: general.

Contents

1	Introduction	3
2	Proposed Solution	3
3	Review of the State of the Art	4
3.1	Random Forests	4
3.2	Decision Trees	4
3.3	Support Vector Machines	5
3.4	Multinomial Regression	6
3.5	Gradient Boosting (GB)	7
3.6	Extreme Gradient Boosting (XGB)	7
4	Dataset	8
4.1	Star Types (Labels)	8
4.2	Features (Attributes)	9
4.3	Size of the Dataset	10
4.4	Class Imbalance	10
5	Experimental Results	10
6	Problems Encountered and Solutions	12
6.1	Overconfidence in Model Predictions	12
6.2	Imbalanced Dataset	12
6.3	Computational Constraints	12
7	Future Prospects	12
8	Conclusion	13
	Appendices	13
A	Project Timeline	13
B	Types of star classes	14

1 Introduction

Classification algorithms play a crucial role in astronomy by enabling the efficient categorization and analysis of vast amounts of data collected from observations. These algorithms help in identifying and classifying celestial objects such as stars, galaxies, and exoplanets based on their properties and behaviors. For instance, they are essential in distinguishing between different types of variable stars, analyzing spectra to determine chemical compositions, and detecting anomalies or rare events. By automating the classification process, astronomers can focus on interpreting results and making new discoveries, ultimately advancing our understanding of the universe. The development of efficient automated classification schemes is becoming of prime importance in astronomy. Large surveys are monitoring millions, and soon billions, of targets. The resulting time series cannot possibly be scrutinized by eye. The identification and study of variable stars require the use of powerful statistical and data mining tools. Automated supervised classification methods provide object type predictions based on the values of a set of attributes characterizing the objects. In the first stage, a collection of prototype objects of known type and attribute values, referred to as the training set, is used to build a model of the dependencies of the types on the attribute values. In the second stage, this model is used to predict the types of other objects of unknown types but with available attribute values. As the naming schemes differ in different publications, we make the following definitions to be used throughout this paper: objects (i.e. stars) are classified into types making use of a number of attributes. We have data on 1,660 stars with various complex attributes obtained by the Hipparcos mission, and we need an automated way to classify them effectively.

2 Proposed Solution

To address the challenge of classifying the 1,660 stars with complex attributes obtained from the Hipparcos mission, we proposed the use of supervised machine learning models. Recognizing the efficacy of these models in handling large datasets with intricate patterns, we focused on evaluating six different algorithms: Random Forest (RF), Decision Trees (DT), Gradient Boosting (GB), Extreme Gradient Boosting (XGB), Support Vector Machines (SVM), and Multinomial Regression.

Our methodology involved a rigorous assessment of each model's performance. We trained and tested the models on the dataset, using accuracy, precision, recall, and confusion matrices as key metrics for evaluation. This approach allowed us to systematically determine which model delivered the highest classification accuracy and the most reliable predictions given the attributes of our dataset.

One of the significant challenges we encountered was the heavily imbalanced nature of the data, with most classes comprising less than 5% of the total dataset. This imbalance can severely impact the performance of machine learning models, leading to biased results. To mitigate this issue, we explored both oversampling and undersampling techniques to balance the data. Oversampling involved increasing the number of instances in the minority classes, while undersampling reduced the number of instances in the majority classes.

After applying these balancing techniques, we re-evaluated the performance of all six models on the balanced datasets. By comparing the results, we aimed to identify which models not only handled the original imbalanced data effectively but also demonstrated

robust performance with balanced data. This comprehensive analysis ensured that the selected model was well-suited for the classification task and capable of delivering accurate and reliable results, thereby facilitating the automated classification of the stars based on their attributes.

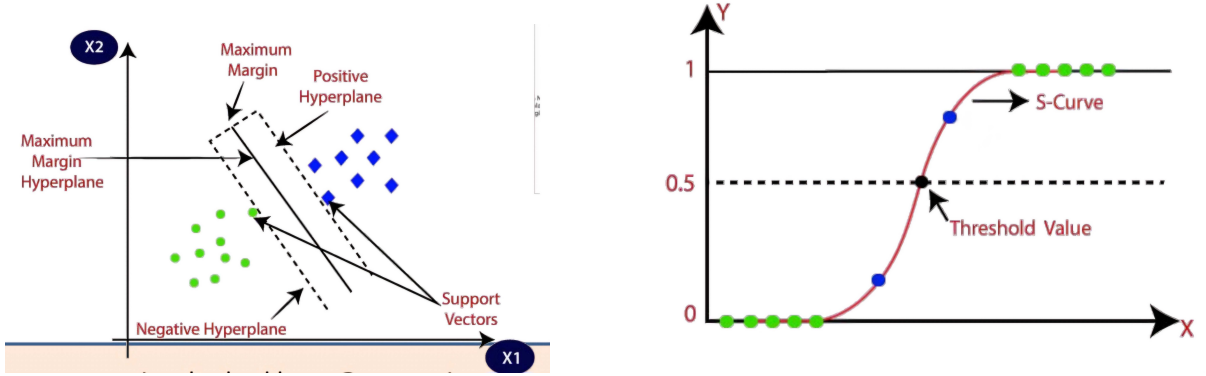


Figure 1: Proposed Solution with highest confidence

3 Review of the State of the Art

3.1 Random Forests

Random forests are an ensemble learning method for classification and regression that operate by constructing a multitude of decision trees during training time and outputting the class that is the mode of the classes (classification) or mean prediction (regression) of the individual trees. The random forest algorithm introduces two main sources of randomness:

1. **Bootstrap Aggregating (Bagging):** Each tree is trained on a random sample of the data, with replacement, leading to varied trees.
2. **Random Feature Selection:** During the construction of each tree, at each split, a random subset of features is selected, and the best split is chosen only from this subset.

Mathematically, let $\{X_1, X_2, \dots, X_n\}$ be the training dataset. For each tree, a bootstrap sample D^* of the data is drawn, resulting in $D^* = \{X_1^*, X_2^*, \dots, X_n^*\}$, where each X_i^* is drawn randomly from $\{X_1, X_2, \dots, X_n\}$ with replacement. At each node of the tree, a subset of m features is selected randomly, and the split that best separates the data according to a certain criterion (e.g., Gini impurity for classification) is chosen.

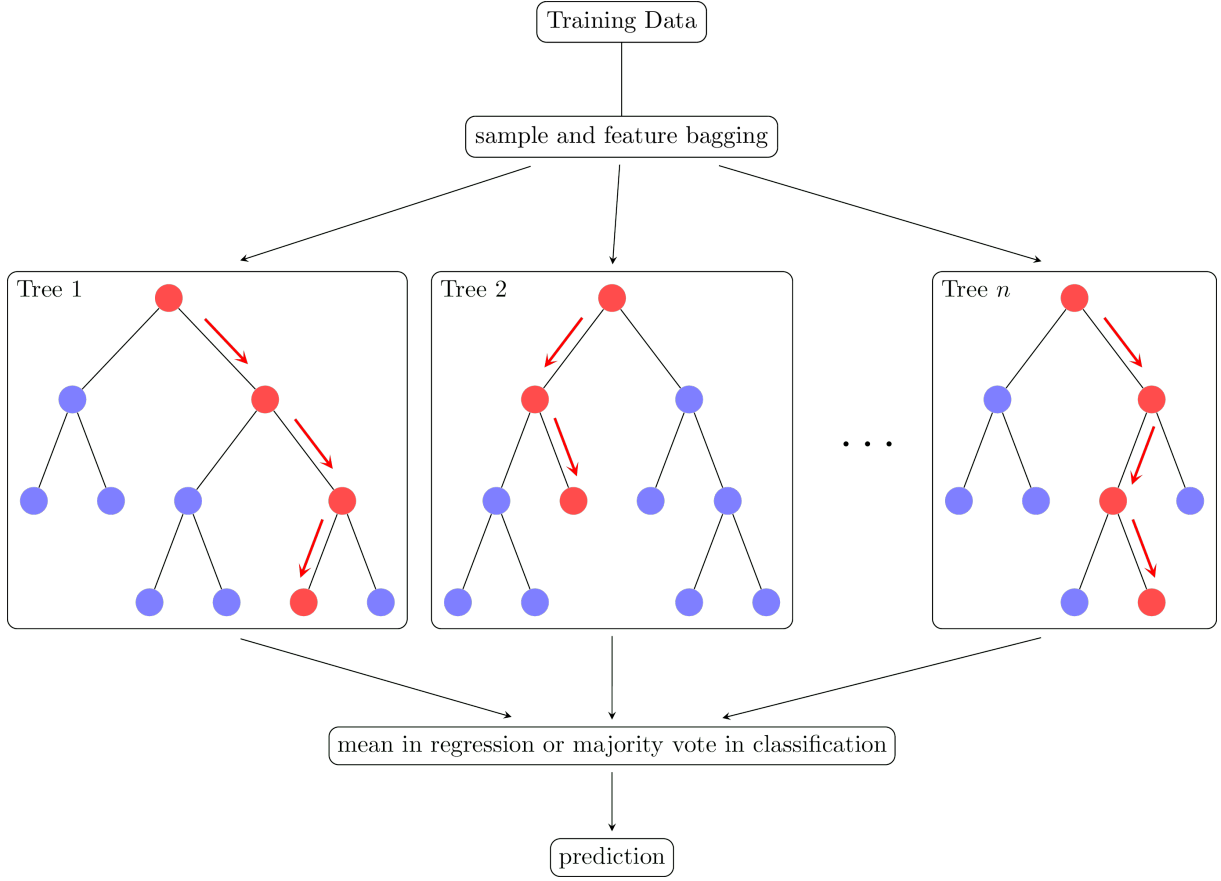
The overall prediction of the random forest for classification is:

$$\hat{y} = \text{mode}(\{h_1(X), h_2(X), \dots, h_B(X)\})$$

where h_i is the prediction of the i -th tree, and B is the total number of trees.

3.2 Decision Trees

A decision tree is a non-parametric supervised learning method used for classification and regression. The goal is to create a model that predicts the value of a target variable by learning simple decision rules inferred from the data features.



A decision tree is composed of nodes, edges, and leaves: - **Nodes**: Test an attribute and represent a condition (e.g., "Is feature X_i less than a threshold?"). - **Edges**: Outcome of a test and connect nodes. - **Leaves**: Terminal nodes that provide the final classification or regression outcome.

The construction of a decision tree involves recursive binary splitting of the data. The splitting criterion can be based on measures such as Gini impurity or information gain (entropy):

$$\text{Gini impurity} = 1 - \sum_{i=1}^C p_i^2$$

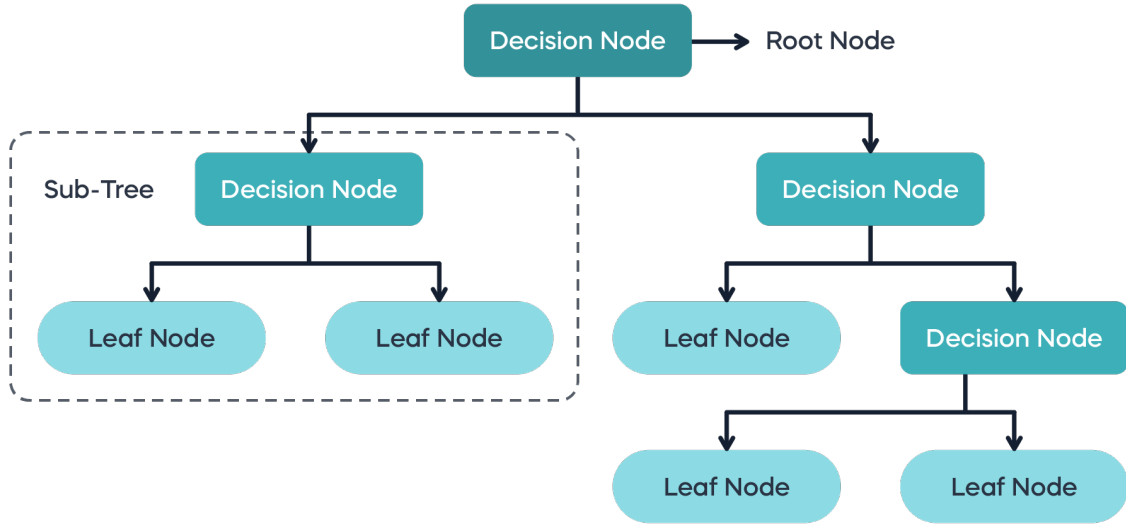
where C is the number of classes and p_i is the proportion of instances of class i in the node.

3.3 Support Vector Machines

Support Vector Machines (SVMs) are supervised learning models used for classification and regression. The objective of the SVM algorithm is to find a hyperplane in an N -dimensional space (N is the number of features) that distinctly classifies the data points.

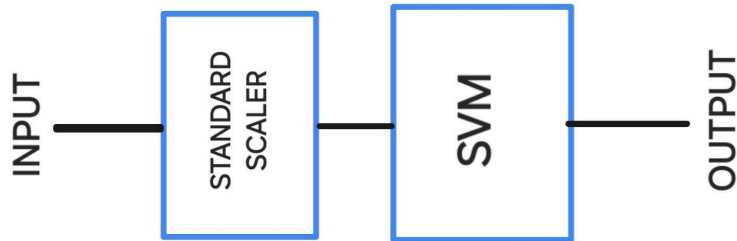
For linearly separable data, the SVM finds the hyperplane that maximizes the margin between the two classes. Mathematically, the optimization problem is:

$$\min_{\mathbf{w}, b} \frac{1}{2} \|\mathbf{w}\|^2$$



$$\text{subject to } y_i(\mathbf{w} \cdot \mathbf{x}_i + b) \geq 1, \forall i$$

For non-linearly separable data, SVMs use kernel functions to map the data into a higher-dimensional space where it becomes linearly separable. Common kernels include: - Linear Kernel: $K(\mathbf{x}, \mathbf{y}) = \mathbf{x} \cdot \mathbf{y}$ - Polynomial Kernel: $K(\mathbf{x}, \mathbf{y}) = (\mathbf{x} \cdot \mathbf{y} + c)^d$ - Radial Basis Function (RBF) Kernel: $K(\mathbf{x}, \mathbf{y}) = \exp(-\gamma \|\mathbf{x} - \mathbf{y}\|^2)$



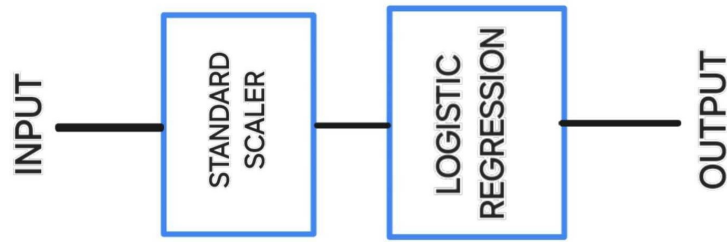
3.4 Multinomial Regression

Multinomial regression is a generalization of logistic regression to multiclass problems, i.e., it is used when the dependent variable is nominal with more than two levels. The model can be expressed as:

$$P(y = k|X) = \frac{\exp(X\beta_k)}{1 + \sum_{j=1}^{K-1} \exp(X\beta_j)}, \quad k = 1, \dots, K-1$$

where $P(y = k|X)$ is the probability of the k -th class given the feature vector X , and β_k is the coefficient vector for the k -th class. The coefficients are estimated by maximizing the likelihood function:

$$L(\beta) = \prod_{i=1}^n P(y_i|X_i)$$



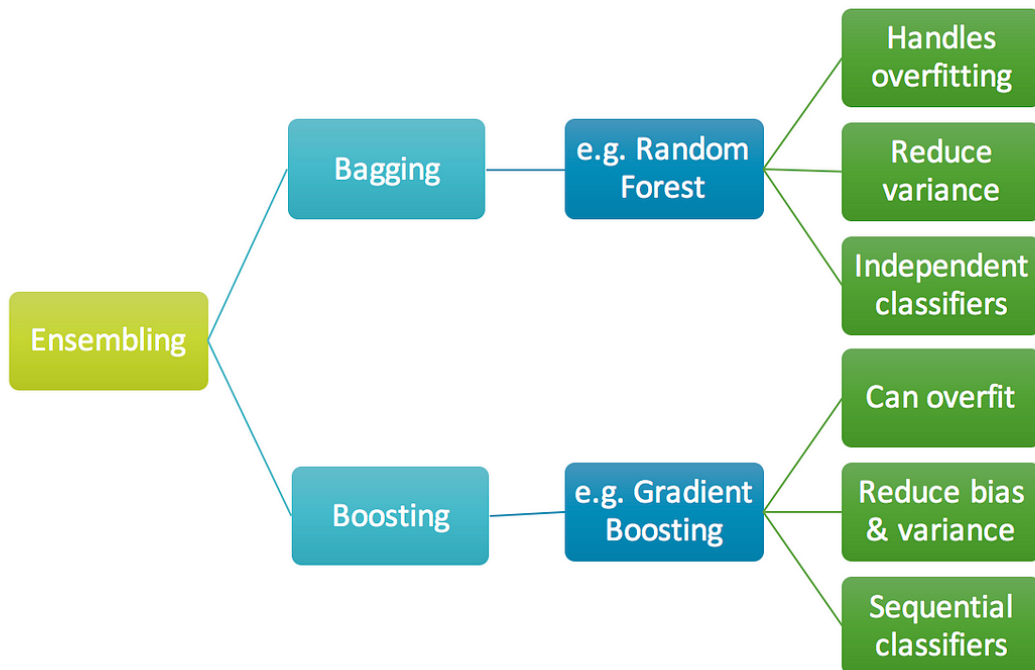
3.5 Gradient Boosting (GB)

Gradient Boosting is an ensemble learning technique that builds models sequentially, each model correcting the errors of its predecessor. For regression, the model is built in a stage-wise fashion, and it generalizes the model by optimizing a cost function.

The prediction at stage m is:

$$\hat{y}_i^{(m)} = \hat{y}_i^{(m-1)} + \nu f_m(X_i)$$

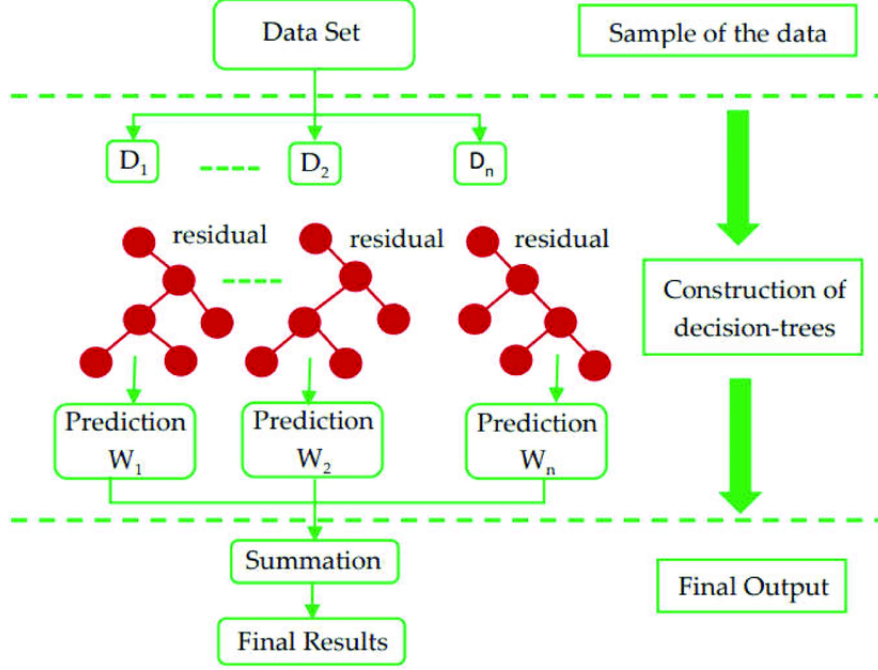
where ν is the learning rate, and f_m is the m -th weak learner, typically a decision tree. The weak learner is trained to minimize the residual errors of the previous model.



3.6 Extreme Gradient Boosting (XGB)

Extreme Gradient Boosting (XGB) is an optimized implementation of the gradient boosting framework designed to be highly efficient, flexible, and portable. It uses more regularized model formalization to control overfitting, which gives it better performance.

The objective function of XGBoost includes a regularization term to penalize the complexity of the model:



$$\mathcal{L}(\theta) = \sum_{i=1}^n l(\hat{y}_i, y_i) + \sum_{k=1}^K \Omega(f_k)$$

where l is a differentiable convex loss function that measures the difference between the prediction $\hat{y}_i$ and the target y_i , and $\Omega(f_k)$ is a regularization term that penalizes the complexity of the model (e.g., the number of leaves in a tree).

4 Dataset

The dataset provided consists of features extracted from observations of variable stars, which vary in brightness over time. These stars are classified into different types based on their variability characteristics.

4.1 Star Types (Labels)

The target variable, **Type**, The dataset consists of 23 different star classes, each representing a unique type of variable star. Eclipsing binaries, which include EA, EB, and EW types, are stars that eclipse one another from our viewpoint, and these are listed with references from the Hipparcos catalogue. Another class, ellipsoidal variables (ELL), refers to stars that exhibit variations in brightness due to their ellipsoidal shape caused by tidal forces from a nearby companion star.

Long-period variables (LPV), which are pulsating stars with long and irregular cycles, are accompanied by subtypes such as RV Tauri and W Virginis, including CWA and CWB subclasses, with data provided by Lebzelter and the AAVSO (American Association of Variable Star Observers). Delta Cepheid (DCEP) variables, which pulsate radially with very regular periods, are also notable in the dataset, with first overtone DCEPS and multi-mode CEP(B) subtypes being represented. Similarly, RR Lyrae stars (RRAB and RRC) exhibit pulsations but are typically shorter-period, older stars.

The dataset also includes Gamma Doradus (GDOR) stars, which are gravity-mode pulsators, and Delta Scuti (DSCT and DSCTC) variables, which pulsate with very short periods. SX Phoenicis (SXPHE) stars, which are similar to Delta Scuti but found in older stellar populations, add to the diversity.

Beta Cephei (BCEP) and Slowly Pulsating B stars (SPB) represent another pulsating category, where Beta Cephei stars pulsate through pressure waves, while SPBs show pulsations through gravity waves. B emission-line stars (BE) display emission lines in their spectra due to the presence of circumstellar gas.

Gamma Cassiopeiae (GCAS) stars exhibit irregular variations in brightness due to mass ejections from their equatorial disks. Alpha Cygni (ACYG) stars, massive supergiants, show irregular variability due to non-radial pulsations. Alpha-2 Canum Venaticorum (ACV) stars have strong magnetic fields and show periodic brightness variations due to spots on their surfaces.

Finally, the SX Arietis (SXARI) class represents stars with rapid rotation and strong magnetic fields, while BY Draconis (BY) and RS Canum Venaticorum (RS) stars are variable due to stellar spots and chromospheric activity. This comprehensive dataset, consisting of 1,661 stars, covers a wide range of stellar variability phenomena across these 23 classes.

Table of types can be referred in appendix section.

4.2 Features (Attributes)

The dataset contains multiple features that help in characterizing the variability of stars. Here are some key features:

- **Log(Period):** The logarithm of the period of the star's brightness variation. Periodic changes in brightness are a key feature in classifying variable stars.
- **Log(Amplitude):** The logarithm of the amplitude of brightness variation. Larger amplitudes indicate more pronounced brightness changes.
- **V-I:** A color index representing the difference in magnitude between the V (visible) and I (infrared) bands, providing information about the star's temperature and age.
- **M_Hipparcos:** The apparent magnitude of the star from the Hipparcos catalog. It's an indicator of the star's brightness as seen from Earth.
- **Scatter:res/raw:** The scatter of residual brightness measurements relative to the raw data. This helps in understanding the variability's regularity or irregularity.
- **Skewness:** A measure of the asymmetry of the brightness variation distribution. Skewness can help distinguish between periodic or irregular variability.
- **Log(1+A2/A1):** A ratio of Fourier harmonics of the star's light curve. This helps to describe the complexity of the variability pattern.
- **P2p scatter:2P/raw:** Point-to-point scatter, comparing the scatter of brightness measurements with a multiple of the star's period.
- **Residual_scatter(%):** The percentage of residual scatter after fitting a model to the brightness variation, giving insight into the irregularities.

- **Phase2 & P2p_slope:** Additional measurements related to the light curve shape and the rate of brightness change over time.

4.3 Size of the Dataset

- **Training Data:** 1,162 instances
- **Test Data:** 498 instances

This split ensures that the model is trained on a substantial portion of the data while leaving enough examples for evaluating the model's performance on unseen data.

4.4 Class Imbalance

As depicted in the pie chart provided earlier, the dataset exhibits class imbalance. For example, the **Other** category makes up about 35.9% of the dataset, while other classes like **EW** or **DCEP** have a much smaller representation. This imbalance can lead to biased model predictions if not handled correctly.

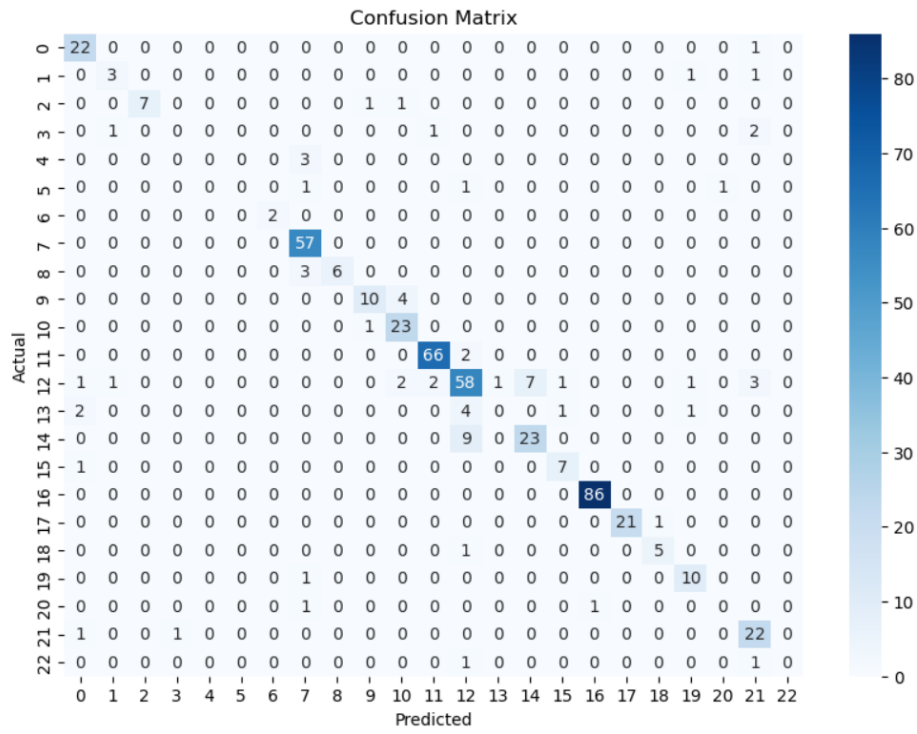
5 Experimental Results

The objective of the experiment was to classify variable stars into different types based on extracted features using machine learning models. We initially cleaned the dataset for outliers and trained six different models, Random Forests, Descision Trees, SVM, Multinomial Regression, Gradient Boost, and Extreme Gradient Boost. The training was done after handling the imbalanced dataset with oversampling techniques. Here's a summary of the results and the challenges encountered during the process.

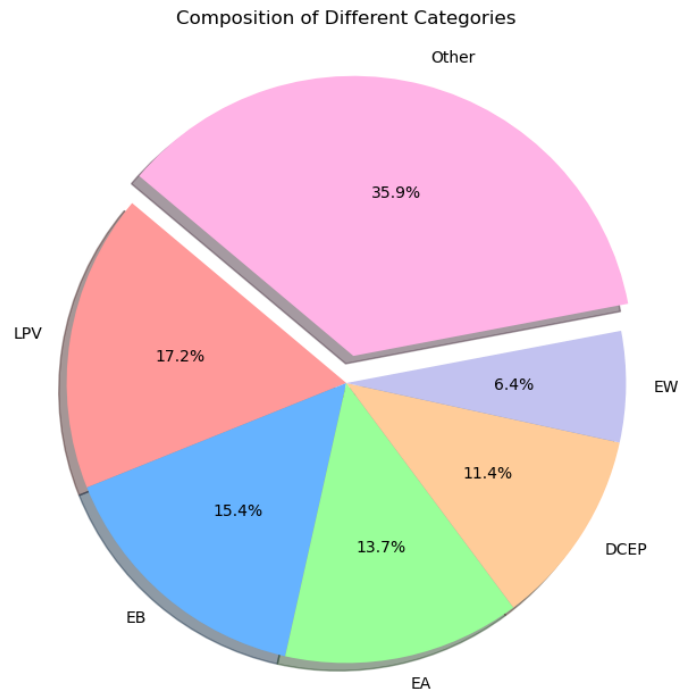
Without balancing:-			
Models ➡ Scores	Random Forest (Vedant)	Support Vector Machines (Rucha)	XGB (Anvit)
Accuracy	Test-0.85, CV-0.84	Test-0.87, CV-0.81	Test-0.8377, CV-0.825
Precision	0.84	0.84	0.8175
Recall	0.85	0.78	0.8377
F1	0.83	0.8	0.8223
With balancing:-			
Models ➡ Scores	Random Forest (Vedant)	Support Vector Machines (Rucha)	XGB (Anvit)
Accuracy	Test-0.86, CV-0.91	Test-0.90, CV-0.93	Test-0.8337, CV-0.816
Precision	0.83	0.87	0.8234
Recall	0.86	0.94	0.8337
F1	0.84	0.9	0.8258

These metrics suggest that the Support Vector Machines model is performing relatively well, with balanced accuracy and precision across different classes.

The confusion matrix (refer to the second image) provides a more detailed breakdown of the predictions per class. For example, the model performed well in classes like 'LPV' (long-period variables) and 'EB' (eclipsing binaries), but struggled with some minority



classes. It also shows that some categories were misclassified due to overlaps in the feature space between certain star types. The pie chart further illustrates the distribution of the



different star categories, with Other taking up the largest share of 35.9%, followed by LPV at 17.2%, and other significant types such as EB, EA, and DCEP.

6 Problems Encountered and Solutions

Throughout the experiment, several challenges were encountered, each requiring specific solutions to improve the model’s performance and ensure the reliability of the results.

6.1 Overconfidence in Model Predictions

During early trials, the model showed tendencies of overconfidence in predicting the majority class (overfitting to majority samples), which led to high precision but low recall for minority classes. This was especially problematic for underrepresented categories like EW and DCEP.

To combat this, we applied techniques like Stratified K-Fold Cross-Validation, which ensured that each fold had a similar class distribution, reducing overfitting to the majority classes. Hyperparameter tuning using GridSearchCV was also employed to select the best combination of parameters that generalize better to unseen data.

6.2 Imbalanced Dataset

The dataset had significant class imbalance, where some star types like ‘Other’ were highly overrepresented (35.9% of the total data), while other classes had very few instances, as seen in the confusion matrix. This imbalance made it difficult for the model to correctly predict minority classes.

We used random oversampling techniques on the training data to address the class imbalance. By replicating the minority class samples, we balanced the dataset to prevent the model from being biased toward the majority classes.

6.3 Computational Constraints

Training the model on a large and complex dataset resulted in long computational times, especially when experimenting with different hyperparameters and model architectures. This limited the number of trials that could be executed within the available time frame.

Summarizing

Addressing these challenges required a combination of data preprocessing techniques, model optimization, and the use of robust validation frameworks to ensure accurate and reliable results

7 Future Prospects

The future of classifying variable stars with machine learning (ML) holds immense potential as astronomical surveys like LSST provide large, high-quality datasets. This will lead to better model accuracy and the discovery of rare star types. Advanced feature engineering, deep learning, and integration of multi-modal data (e.g., photometric and spectroscopic) will enhance classification precision.

Explainable AI (XAI) will become crucial for interpreting ML model decisions, ensuring scientific validity. Real-time classification and anomaly detection can identify rare stellar events as they occur. Transfer learning and domain adaptation will allow models to generalize across different instruments.

8 Conclusion

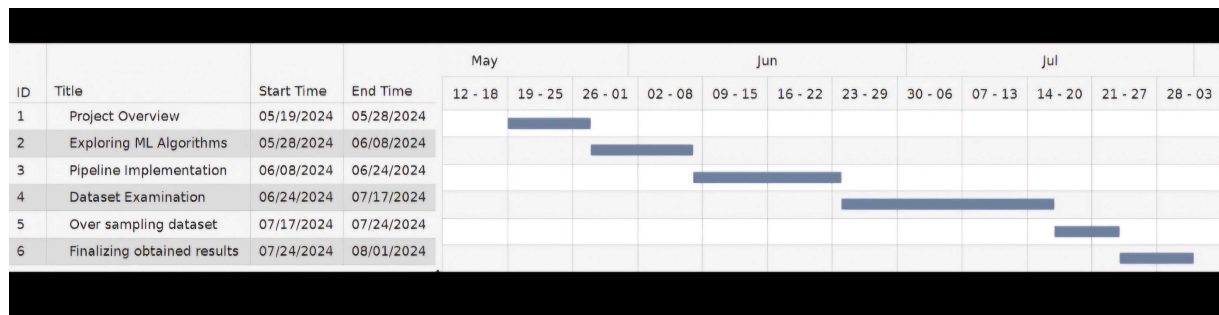
Our project has successfully demonstrated that machine learning algorithms, especially those involving supervised learning techniques, can efficiently classify variable stars into their respective types with high accuracy. We have obtained robust results that not only improve our understanding of the behavior of variable stars but also contribute to the broader astronomical research community. The model and its performance show significant potential for automating the classification process, which could greatly facilitate the analysis of large-scale astronomical surveys. Overall, this project highlights the value of integrating machine learning techniques into astrophysical research, paving the way for more detailed and efficient studies of variables stars. [7] [1] [2] [3] [4] [5] [6]

References

- [1] <https://kaggle.com/datasets/87e3c68116789b2e5a5a70abdedf50a05272ba5673244e8d43c8009ee0e7d>
- [2] <https://scikit-learn.org/stable/modules/generated/sklearn.pipeline.Pipeline.html>.
- [3] https://scikit-learn.org/stable/common_pitfalls.html.
- [4] <https://www.youtube.com/watch?v=8MxDgHMJnic>.
- [5] <https://www.youtube.com/watch?v=HZ9MUzCRlzl>.
- [6] https://en.m.wikipedia.org/wiki/Machine_learning.
- [7] P Dubath, L Rimoldini, M Süveges, Jonas Blomme, M López, LM Sarro, Joris De Ridder, J Cuypers, L Guy, I Lecoœur, et al. Random forest automated supervised classification of hipparcos periodic variable stars. *Monthly Notices of the Royal Astronomical Society*, 414(3):2602–2617, 2011.

Appendices

A Project Timeline



B Types of star classes

Table 1: Variable Star Types and References

Type	Num	Main reference
Eclipsing binary		
EA	228	<i>Hipparcos</i>
EB	255	<i>Hipparcos</i>
EW	107	<i>Hipparcos</i>
Ellipsoidal		
ELL	27	<i>Hipparcos</i>
Long period variable		
LPV	285	Lebzelter (p.c.)
RV Tauri	5	AAVSO
W Virginis		
CWA	9	AAVSO
CWB	6	AAVSO
Delta Cepheid		
DCEP	189	AAVSO
DCEPS (first overtone)	31	AAVSO
CEP(B) (multi-mode)	11	AAVSO
RR Lyrae		
RRAB	72	AAVSO
RRC	10	AAVSO
Gamma Doradus		
GDOR	27	De Cat (p.c.)
Delta Scuti		
DSCT	43	AAVSO
DSCTC (low amplitude)	81	AAVSO
SX Phoenicis		
SXPHE	4	AAVSO
Beta Cephei		
BCEP	30	De Cat (p.c.)
SPB (Slowly Pulsating B star)	81	De Cat (p.c.)
BE (B emission-line star)	9	AAVSO
Gamma Cassiopeiae		
GCAS	7	AAVSO
Alpha Cygni		
ACYG	18	AAVSO
Alpha-2 Canum Venaticorum		
ACV	77	Romanyuk (p.c.)
SX Arietis		
SXARI	7	Romanyuk (p.c.)
BY Draconis		
BY	5	Eker et al. (2008)
RS Canum Venaticorum		
RS	30	Eker et al. (2008)
Total	1661	

AAVSO: Watson et al. (2010).